QMSS 5070

GIS and Spatial Analysis

Fall 2026
Wednesdays, 12:10 PM – 2:00 PM
Pupin Laboratories 214

Instructor

Name: Edwin Grimsley, Ph.D.

Office Hours: Location TBD; Wednesdays before/after class, or via Calendly appointment

Virtual Office Hours: Via Zoom by appointment — email to schedule a time and receive the link (https://columbiauniversity.zoom.us/j/2400680218)

Email (primary): eg3033@columbia.edu

Email (secondary): edwin.grimsley@baruch.cuny.edu

Teaching Assistants

Manas Agnihotri — maa2416@columbia.edu

Shreya Shetty — svs2148@columbia.edu

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I. Course Description

The course introduces map-making skills, utilization of spatial data, and spatial analysis in policy applications. It emphasizes the interdisciplinary nature of Geographic Information Systems (GIS), which have become an increasingly helpful tool for observing and analyzing social and physical phenomena over space. Most of the physical and social sciences, as well as multiple professions, are expanding the infrastructure of GIS data that allows for new interdisciplinary analyses to occur and for unexamined potential spatial relationships to be uncovered. This is especially true for social scientific analyses, where a wealth of spatial data on cultural, economic, physical, and demographic characteristics remains unexamined but is easily accessible from many sources. It covers introductory concepts and tools related to Geographic Information Systems (GIS), including spatial data acquisition, gathering and analyzing Census population data, spatial data management, and spatial data analysis.

We will cover GIS concepts such as spatial data acquisition, Census data analysis, spatial data management, and spatial analysis. A primary goal of this course is to provide a non-threatening introduction to GIS and its relation to hypothesis testing using statistical software in R. Students will learn to create and interpret thematic maps via hands-on experience with QGIS. Advanced topics include spatial data construction, quantitative applications of spatial data, and critical GIS practices related to social and racial justice.

Through homework assignments, lab exercises, and a research project of your choice, we will obtain hands-on experience with the kinds of GIS and spatial statistical methods most frequently employed by social scientists. Most importantly, the course will provide students a basic grasp of what spatial data analysis is all about.

Prerequisites: No previous GIS experience is required; however, at least one prior course in statistics (preferably including linear regression) is recommended.

II. Course Objectives

- **Develop proficiency** in GIS software for spatial data analysis.
- **Understand theoretical foundations** of spatial analysis.
- **Apply GIS tools** to real-world public policy and social science problems.
- **Create effective maps and visualizations** to communicate spatial data.

Class sessions will include:

- **Lectures** that introduce key GIS concepts, tools, and theory.
- **Lab exercises** during class to reinforce hands-on skills.
- **Homework assignments** posted on Courseworks, building cumulative knowledge.
- **Midterm exam** to assess foundational skills and knowledge.
- **Final paper** serving as the culminating project, where you apply GIS methods to a topic of your choice.
- **Short research presentation** to showcase your final paper's findings.

Because the material is cumulative and lectures expand upon rather than repeat the readings, **regular attendance and keeping current with assignments are essential**.

III. Requirements and Assignments

There will be eight homework assignments; written work (1–2 pages, double-spaced, plus output) is due at the start of class on Tuesday of the following week. Homework assignments in the latter half of the course will center on completion of a small research project in the student's area of interest using geographically referenced data.

You will learn by doing. In-class labs require the use of a laptop and the various software tools we use in the course. Please bring a laptop to each class if you are attending in person. Because the content of this course is cumulative and lectures expand upon rather than repeat the readings, it is essential that you attend class and do not fall behind.

1. Homework Assignments (50%)

- **Eight** total assignments (lowest score dropped).
- Each due by the start of class on Tuesday the following week.
- Later assignments will guide you toward developing your final research paper.

2. Midterm Exam (20%)

- Assesses foundational GIS concepts, spatial data structures, and basic mapping techniques.
- May include short answers, GIS data interpretation, and hands-on exercises.
- Preparation should involve reviewing lecture notes, readings, and completing practice exercises.

3. Final Paper (15%)

- Acts as the **culminating project** for the course, in lieu of a traditional final exam.
- You will choose a topic of interest involving spatial data, apply GIS and analytical methods, and present findings in a professionally formatted paper.
- Further guidelines regarding length, structure, and citation format will be posted on Courseworks.

4. Research Presentation (5%)

- A short in-class presentation of your final paper's findings.
- Focus on key insights and demonstrate your GIS mapping/analysis skills.

5. Attendance & Class Participation (10%)

- Required for every class meeting.
- Peer reviews on research project.
- Some sessions include lab exercises that count toward participation.
- If you cannot attend due to unforeseen circumstances, inform the instructor in advance.

IV. Grading Breakdown

Assignment	Percentage
Homework Assignments (8 total, lowest dropped)	45%
Midterm Exam	25%
Final Paper	15%
Research Presentation	5%
Attendance / Class Participation	10%
Total	**100%**

Note: the percentages above (from the grading table) differ slightly from those listed in Section III (50/20/15/5/10). You may want to confirm the correct weights with the instructor and update whichever section is outdated.

- **Extra Credit:** There may be limited extra credit opportunities later in the semester.

V. Course Policies

Late Work

Late homework will only be accepted under special circumstances and with prior instructor approval. Since content builds weekly, it is crucial to maintain the pace.

Courseworks

All lectures, assignments, and announcements will be posted on Courseworks. Check frequently for updates and materials.

Academic Integrity

Maintain the highest standards of honesty. Plagiarism or cheating in any form will not be tolerated and may result in failing the course.

Expectation of Regular Participation and Utilization of Courseworks

We will be monitoring student participation and completion of assignments throughout the semester. We want to make sure that students are consistently engaged, and if that becomes difficult, that students alert us to their situations.

VI. Software

- **QGIS** (primary open-source GIS software)
- **ArcGIS Pro** (optional, if provided a license from QMSS)
- **GeoDa** (for exploratory spatial data analysis)
- **R** (for statistical tests and spatial packages)

VII. Required Books

(Check eBook availability from the Columbia University Library before buying.) This course uses open-source materials — both required texts are available to students in full PDF or readable form.

1. Essentials of Geographic Information Systems

Authors: Jonathan E. Campbell & Michael Shin

Edition: Version 1.0 (open textbook / online edition)

Publisher: Saylor Foundation (via Open Textbook Library / Saylor Academy)

Year: 2011/2012 (online open edition)

Link: saylordotorg.github.io/text_essentials-of-geographic-information-systems

2. The SAGE Handbook of Spatial Analysis

Fully available as a PDF through the Columbia University Library.

Editors: A. Stewart Fotheringham & Peter A. Rogerson

Publisher: SAGE Publications

Year: 2008/2009

ISBN: 978-0857020130

Recommended (Not Required)

- Steinberg, S. J. & Steinberg, S. L. 2006. *GIS: Geographic Information Systems for the Social Sciences.* Thousand Oaks, CA: Sage.
- Fotheringham, A. S., Brunsdon, C., & Charlton, M. 2000. *Quantitative Geography: Perspectives on Spatial Data Analysis.* London: Sage.
- Mitchell, A. 1999. *The ESRI Guide to GIS Analysis, Volume 1: Geographic Patterns and Relationships.* Redlands, CA: ESRI Press.
- Mitchell, A. 2005. *The ESRI Guide to GIS Analysis, Volume 2: Spatial Measurements and Statistics.* Redlands, CA: ESRI Press.
- Lloyd, C. D. 2010. *Spatial Data Analysis: An Introduction for GIS Users.* Oxford: Oxford University Press.
- Howell, F. M., Porter, J. R., & Matthews, S. 2016. *Recapturing Space: New Middle Range Theory in Spatial Demography.* Springer. (eBook available via Columbia Library.)
- Porter, J. R. & Howell, F. M. 2012. *Geographical Sociology.* Springer. (eBook available via Columbia Library.)
- Ormsby, T., Napoleon, E., Burke, R., Groessl, C., & Bowden, L. 2010. *Getting to Know ArcGIS Desktop, 2nd Ed., Updated for ArcGIS 10.* Redlands, CA: ESRI Press.
- Allen, D. W. 2010. *GIS Tutorial 2: Spatial Analysis Workbook, 2nd Ed.* Redlands, CA: ESRI Press.
- Monmonier, M. 1993. *Mapping It Out: Expository Cartography for the Humanities and Social Sciences.* Chicago: University of Chicago Press.

Related Articles (Free Download)

- Anselin, L. 1989. "What Is Special about Spatial Data? Alternative Perspectives on Spatial Data Analysis." Working Paper. Santa Barbara, CA: NCGIA.
- Anselin, L. 1995. "Local Indicators of Spatial Association — LISA." *Geographical Analysis* 27(2): 93–115.
- Anselin, L. 2002. "Under the Hood: Issues in the Specification and Interpretation of Spatial Regression Models." *Agricultural Economics* 27(3): 247–267.

- Anselin, L. & Syabri, I. 2006. "GeoDa: An Introduction to Spatial Data Analysis." *Geographical Analysis* 38(1): 5–22.
- Haining, R. 2009. "The Special Nature of Spatial Data." In Fotheringham & Rogerson, eds., *The SAGE Handbook of Spatial Analysis.* London: Sage.
- Cho, W. K. T. & Gimpel, J. G. 2012. "Geographic Information Systems and the Spatial Dimensions of American Politics." *Annual Review of Political Science*, 443–460.
- Gleditsch, K. S. & Weidmann, N. B. 2012. "Richardson in the Information Age: Geographic Information Systems and Spatial Data in International Studies." *Annual Review of Political Science*, 461–482.

Software Manuals and Tutorials (Free Download)

- Levine, N. 2004. *CrimeStat III: A Spatial Statistics Program for the Analysis of Crime Incident Locations.* Houston, TX: Ned Levine and Associates / Washington, DC: National Institute of Justice.
- Anselin, L. 2003. *GeoDa 0.9 User's Guide.* Spatial Analysis Laboratory, University of Illinois.
- Anselin, L. 2003. *GeoDa 0.9.5-I Release Notes.* Spatial Analysis Laboratory, University of Illinois.

Useful websites (visited throughout the semester for readings and lab materials): Center for Spatially Integrated Social Science (csiss.org); GeoDa Center, Arizona State University.

Important Notes

1. Homeworks

- **8 total** homework assignments; your **lowest score** is dropped.
- Each due the **following Tuesday** at 11:59 PM unless otherwise noted.
- **No late submissions** without prior approval.

2. Midterm

- Handed out in **Week 7**, due **Week 8**.
- Covers foundational GIS and spatial analysis concepts from Weeks 1–6.

3. Final Paper & Presentation

- In lieu of a final exam, you will produce a **research paper** applying spatial methods learned in class.
- Short **in-class presentation** in Weeks 12–13.
- Detailed guidelines will be posted on Courseworks.

4. Participation & Attendance

- Regular attendance is essential, as labs reinforce lecture and readings.
- Please inform the instructor **in advance** if you must miss class.

5. Readings & Software

- Primary software: **QGIS**, **ArcGIS**, **GeoDa**, **R** (Leaflet, etc.).
- Supplementary Python scripts may be introduced but are optional.
- All essential readings and lab materials will be posted on Courseworks or accessible online.

VIII. Schedule

Week 1 (September 9): Orientation to Spatial Thinking & What Is GIS?

Topics

- Course administration & expectations
- What is GIS? Core components and history
- Why spatial thinking matters for policy & social science

Readings

- Essentials of Geographic Information Systems — Ch. 1: Introduction to Geographic Information Systems
- The SAGE Handbook of Spatial Analysis — Ch. 1 (Fotheringham & Rogerson): The Foundations of Spatial Analysis
- Andy Mitchell, The ESRI Guide to GIS Analysis, Vol. 1 — Ch. 1: Introducing GIS Analysis (optional)

Homework

- HW #1 assigned: *Mapping Your Journey by Hand* — due Week 2 (start of class)

Week 2 (September 16): Making Maps — Basic Visualization

Topics

- Thematic mapping
- Classification schemes
- Color, symbols, scale, legends
- Ethical cartography

Readings

- Essentials of GIS — Ch. 2: Map Anatomy and Geographic Data; Section 6.3: Data Classification
- Andy Mitchell, ESRI Guide to GIS Analysis, Vol. 1 — Ch. 2: Mapping Where Things Are; Ch. 3: Mapping the Most and Least

Class Assignment

- Lab #1 — QGIS: Choropleths, symbols, labels

Homework

- HW #1 due
- HW #2 assigned: Mapping Visualization in the U.S. — due Week 3

Week 3 (September 23): Exploring Spatial Data I

Topics

- Census geography & data structures
- MAUP, scale, ecological fallacy
- Tabular data & joins

Readings

- Essentials of GIS — Ch. 3: Data, Information, and Where to Find Them
- U.S. Census Bureau — Understanding and Using American Community Survey Data: What All Data Users Need to Know
- Andy Mitchell, ESRI Guide to GIS Analysis, Vol. 1 — Ch. 3 (review); Ch. 4: Mapping Density

Class Assignment

- Lab #2

Homework

- HW #2 due
- HW #3 assigned — due Week 4

Week 4 (September 30): Exploring Spatial Data II

Topics

- Geoprocessing
- Buffers, overlays, points-in-polygons
- Change over time

Readings

- Essentials of GIS — Ch. 7: Analytical Operations in GIS

Class Assignment

- Lab #3 — Buffer & overlay exercises

Homework

- HW #3 due

Week 5 (October 7): Advanced Data Wrangling, CRS & Geocoding

Topics

- CRS and projections
- Geocoding & address matching
- Data cleaning

Readings

- Essentials of GIS — Ch. 2 (skim); Sections 6.1–6.2: Data Characteristics and Visualization; Ch. 7: Geospatial Analysis I: Vector Operations

Class Assignment

- Lab #4 — Geocoding lab

Homework

- HW #4 assigned — due Week 6

Week 6 (October 14): ESDA, Point Patterns & Hotspots

Topics

- Spatial randomness
- Point pattern analysis
- Hotspot detection

Readings

- Essentials of GIS — Ch. 12: Spatial Statistics
- SAGE Handbook — Ch. 21: Exploratory Spatial Data Analysis; Ch. 22: Point Pattern Analysis

Class Assignment

- Lab #5

Homework

- HW #4 due
- HW #5 assigned

Week 7 (October 21): R for Spatial Data & Interactive Mapping (midterm handed out)

Topics

- R spatial workflows
- Interactive maps
- Integrating QGIS + R

Readings

- Essentials of GIS — Ch. 13: GIS and the Internet
- SAGE Handbook — Ch. 28: Web-Based Spatial Analysis

Class Assignment

- Lab #6 — no submission

Homework

- Take-home midterm distributed

Week 8 (October 28): Midterm Due & Research Design

Topics

- Midterm due (Friday, October 30)
- Research project design
- Data & methods planning

Homework

- HW #6 assigned: Research Questions & Data

Week 9 (November 4): Spatial Autocorrelation

Topics

- Global & local spatial autocorrelation
- Weight matrices

Readings

- Essentials of GIS — Ch. 14: Advanced Spatial Statistics
- SAGE Handbook — Ch. 23: Spatial Autocorrelation; Ch. 24: Local Indicators of Spatial Association

Class Assignment

- Lab #7

Homework

- HW #6 due
- HW #7 assigned — due Week 10

Week 10 (November 11): Regression & Spatial Dependence

Topics

- OLS diagnostics
- Spatial lag & error models

Readings

- Essentials of GIS — Ch. 15: Spatial Regression
- SAGE Handbook — Ch. 26: Spatial Regression Modeling

Class Assignment

- Lab #8

Homework

- HW #7 due
- HW #8 assigned

Week 11 (November 18): Advanced Spatial Models

Topics

- Spatial Durbin
- GWR
- Intro to spatial ML

Readings

- SAGE Handbook — Ch. 27: Geographically Weighted Regression; Ch. 29: Emerging Methods in Spatial Analysis

Class Assignment

- Lab #9

Homework

- HW #8 due

November 25 — No Class

Weeks 12–13 (December 2 and December 9): Final Presentations

Week 15 (December 20, Sunday): Final Paper Due

Letter Grades

Student grades will be assigned according to the following criteria:

Grade	Range	Description
A	94–100	Excellent — exceptional, professional-quality work: thorough, well-reasoned, creative, and methodologically sophisticated.
A-	90–93	Very good — strong, creative, thorough work showing solid command of appropriate methods; meets professional standards.
B+	87–89	Good — sound, thorough, methodologically solid work that fully accomplishes the course's basic objectives.
B	84–86	Adequate — competent work with some evident weaknesses; understanding of some issues is less than complete.
B-	80–83	Borderline — meets only minimal graduate expectations; understanding and methods are minimally adequate.
C+/C/C-	70–79	Deficient — does not meet minimal graduate expectations; numerous errors or gaps in understanding.
F	< 70	Fail — does not meet minimal expectations for course credit; weaknesses are pervasive.